

Decomposed Velocity Fields During Grokking

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Abstract

Grokking refers to the phenomenon of neural networks hitting near-perfect training accuracy early and then taking orders of magnitude more steps to generalise on a held-out dataset [7]. The primary contribution is a pointwise velocity-field Hodge decomposition across eight grokking runs. Across eight training seeds, the baseline mean exact, coexact, and harmonic fractions change from (0.325, 0.450, 0.225) during the fixed transition window to (0.473, 0.331, 0.196) post-grokking. At the baseline setting, the post-minus-transition change in coexact-minus-exact fraction is -0.267 (95% seed-level t interval $[-0.400, -0.135]$). Its direction is the same in all 56 correlated seed-by-setting sensitivity comparisons and survives seed-relative phase alignment (-0.360 , 95% interval $[-0.487, -0.234]$). Permuting point correspondence removes the post-grokking exact enrichment. Effective dimensionality and resolvent BW distance provide global corroborating diagnostics. The robust result is therefore a post-grokking shift toward exact velocity, rather than a universal coexact signature during the transition itself.

1 Introduction

Grokking refers to the phenomenon of neural networks hitting near-perfect training accuracy early and then taking orders of magnitude more steps to generalise on a held-out dataset [7]. The delay arises not from a lack of capacity but from the time the network takes to find a representation that generalises from data it can already fit.

The network first memorises the training set as a lookup table, then gradually finds a compact algorithmic solution. For modular addition $(a + b) \bmod P$ with P prime, Nanda et al. [5] used mechanistic interpretability to show that the generalising solution is a Fourier representation. The solution learned embeds each residue class $a \in \mathbb{Z}/P\mathbb{Z}$ as a point on a circle in a 2D subspace, with the angular position of a being $2\pi ka/P$ for some frequency k . Addition then follows from the trigonometric identity

$$\cos\left(\frac{2\pi k(a+b)}{P}\right) = \cos\left(\frac{2\pi ka}{P}\right) \cos\left(\frac{2\pi kb}{P}\right) - \sin\left(\frac{2\pi ka}{P}\right) \sin\left(\frac{2\pi kb}{P}\right). \quad (1)$$

This clock algorithm needs only a small number of Fourier frequencies. It places the P residue classes on a low-dimensional manifold with the topology of a circle for one frequency or a torus for several.

The primary contribution is a pointwise velocity-field Hodge decomposition across eight grokking runs. The transition is clearly visible in the representation and in the global corroborating diagnostics, while the completed seed, probe, parameter, synthetic-field, and correspondence-null checks delimit the Hodge claim. Supplementary geometric diagnostics and an audit of the evidential status of these claims appear in the appendix.

In this paper we study the grokking transition through a primary spatial analysis and complementary global diagnostics. Both are built on a shared representation space made from activation checkpoints.

1. **Primary spatial analysis.** The probe displacement field $v_i(t) = x_i(t + \Delta t) - x_i(t)$ between consecutive checkpoints is a vector field on the point cloud at time t . We treat it as a degree-1 object and perform Hodge decomposition via DiffusionGeometry, partitioning the velocity into exact (gradient), coexact (curl), and harmonic components.
2. **Global corroborating diagnostics.** We track the effective dimension of the centred activation covariance. At each checkpoint we also build the graph Laplacian L_t and derive the regularised inverse, or resolvent covariance, $\Sigma_t = (L_t + \varepsilon I)^{-1}$ on the retained non-trivial spectral subspace, an inversion which emphasises low-frequency structure. We measure the Bures–Wasserstein distance between consecutive Gaussians.

The spatial analysis captures what the representation is *doing*, by tracking how activation vectors move between checkpoints. The global diagnostics capture how the representation and its diffusion geometry evolve.

2 Experimental Setup

2.1 Architecture and Training

The training setup follows Nanda et al. [5] verbatim except for the single-head reduction discussed below; we record the parameters here for completeness.

We use a deliberately simplified single-head variant of the Nanda et al. [5] architecture:

- Task: modular addition $(a + b) \bmod 113$, with $P = 113$ (prime).
- Architecture: 1-layer transformer (TransformerLens 2.17.0), embedding dimension $d_{\text{model}} = 128$, single attention head ($d_{\text{head}} = 128$), MLP hidden dimension 512, no layer normalisation.
- Optimiser: AdamW with $\beta = (0.9, 0.98)$, weight decay 1.0.
- Training: 25,000 epochs, full-batch.
- Data split: 70% of all (a, b) pairs held out for validation.
- Reproducibility: for each of eight runs, the data seed is also supplied to `torch.manual_seed` before model creation; there is no separate model seed. The runs use seeds 598, 599, 777, 1001, 2027, 3141, 4096, and 8080, and are pinned to TransformerLens 2.17.0.

We intentionally did not aim for an exact reproduction of the four-head model in Nanda et al. [5]. We reduced to a single head because the four-head version gave a noisier representation. In that version the circular correlation of the recovered coordinates falls from ≈ 0.85 to ≈ 0.44 , which makes the Fourier analysis we run later quite difficult. Indeed, the 1-head variant shows the same memorisation, transition and generalisation pattern that we want to analyse, and we found the Fourier geometry easier to read.

At 51 checkpoints (every 500 epochs plus initialisation) we extract the MLP hidden activations (post-ReLU, $n_{\text{hidden}} = 512$) at the “=” token position for a fixed probe set of $n = 500$ subsampled input pairs. This gives a sequence of point clouds $X_t \in \mathbb{R}^{500 \times 512}$.

2.2 Methods

Spatial side: DG velocity Hodge decomposition. Given consecutive checkpoints X_t and $X_{t+\Delta t}$, the probe displacement field is

$$v_i(t) = x_i(t + \Delta t) - x_i(t).$$

We reduce X_t to 10 dimensions with PCA. We also project v_i through the same basis. We then build a DiffusionGeometry model [3, 4] with $k\text{NN} = 15$ and 50 spectral basis functions on the reduced point cloud. We promote the displacement to a degree-1 form and apply the Hodge decomposition

$$\omega = df + \delta g + h,$$

where df is exact (the gradient of a 0-form), δg is coexact (the codifferential of a 2-form), and h is harmonic. The three components are orthogonal on a smooth manifold. On a finite point cloud the numerical cross terms are small but non-zero. The raw magnitude charts are very noisy, so to interpret them we look at the composition rather than the raw magnitudes. We report normalised energy fractions ($\hat{e}_{\text{exact}}, \hat{e}_{\text{coexact}}, \hat{e}_{\text{harmonic}}$) that sum to one by construction, telling us how much of the probes’ motion is gradient-like, coexact-like, or left in the residual harmonic component. We read the harmonic term conservatively in this paper: without an independent persistent homology or Betti number calculation we cannot treat the harmonic energy as direct evidence for a topological cycle.

For the robustness analysis we use eight independently trained seeds, four random 400-example probe subsets, and the same nine preselected checkpoint pairs. Starting from the baseline setting of PCA dimension 10, $k\text{NN} = 15$, and 50 basis functions, we vary one setting at a time over PCA dimensions 5 and 20, $k\text{NN}$ values 10 and 20, and 30 and 75 basis functions. This gives seven settings and 56 correlated seed-by-setting sensitivity comparisons; the eight training seeds, rather than those 56 comparisons, are the independent units for inference. The primary fixed convention labels pair midpoints before 1,500 as pre-grokking, 1,500–4,000 inclusive as transition, and later midpoints as post-grokking. As a phase-definition sensitivity check, we anchor each seed at its first saved checkpoint with validation accuracy at least 0.5, define its transition as $[-1,500, +1,000]$ epochs around that event, and reclassify the same nine pairs (yielding two to five pairs per phase, depending on seed). At the baseline parameter setting, we draw one separate destination-point permutation for each seed, probe subset, and checkpoint pair: $8 \times 4 \times 9 = 288$ permutations. Actual and null fractions are paired within the same training seed, probe subset, and checkpoint pair, averaged within seed, and then analysed across the eight seeds. All reported 95% intervals are two-sided Student t intervals over eight seed-level means (7 degrees of freedom). Finally, we calibrate the implementation on synthetic gradient, rotational, and discrete harmonic fields.

Operator side: resolvent BW distance. At each checkpoint we build the symmetric normalised graph Laplacian L_t (adaptive-bandwidth Gaussian kernel, $k\text{NN} = 15$) on the PCA-reduced probe cloud [2] and extract $K = 30$ non-trivial eigenpairs, excluding the constant mode. The resolvent covariance is $\Sigma_t = (L_t + \varepsilon I)^{-1}$ on this retained non-trivial spectrum, with $\varepsilon = 0.01$, which maps small non-trivial Laplacian eigenvalues to large covariance eigenvalues and so emphasises low-frequency structure. We compute the BW distance $d_{\text{BW}}(\Sigma_t, \Sigma_{t+\Delta t})$ between consecutive checkpoints [1].

Given two geometric states as Gaussians $\mathcal{N}(0, \Sigma_0)$ and $\mathcal{N}(0, \Sigma_1)$ where $\Sigma_t = f(L_t)$, the Bures–Wasserstein distance [1] is:

$$d_{\text{BW}}^2(\Sigma_0, \Sigma_1) = \text{tr}(\Sigma_0) + \text{tr}(\Sigma_1) - 2 \text{tr} \left[\left(\Sigma_0^{1/2} \Sigma_1 \Sigma_0^{1/2} \right)^{1/2} \right]. \quad (2)$$

This equals the Wasserstein-2 distance $W_2(\mathcal{N}(0, \Sigma_0), \mathcal{N}(0, \Sigma_1))$. Each Gaussian encodes the diffusion geometry of its manifold, and the BW distance is the cost of morphing one geometry into the other.

2.3 Scope and Limitations

The quantitative Hodge analysis within this paper uses eight independently trained seeds. We also ran probe robustness checks on the Hodge fractions and on the global corroborating diagnostics. Across four random 400-example subsets, the Hodge phase shift, dimensional collapse, and BW spiking at the transition point survive.

The Hodge fractions remain sensitive to PCA dimension, neighbourhood range, and spectral cutoff. The completed one-at-a-time sweep supports the direction of the post-grokking shift, but not universal coexact dominance during the transition or universal exact dominance after it. The eight runs reach 99% validation accuracy between epochs 3,500 and 5,500; the directional result survives both the fixed and seed-relative phase definitions, although neither exhausts reasonable phase conventions. The conclusions therefore remain bound to this architecture, task, and finite parameter grid.

3 Results

3.1 Training Dynamics

A few hundred rounds into training the model hits perfect training accuracy. Yet when tested on unseen data, performance barely beats random guessing until epoch 1,500. From that point forward improvement accelerates past the tipping point, reaching 94.3% validation accuracy by epoch 3,000 and 100% by epoch 4,000.

Later analyses tie back to this timeline as their foundation. Training divides into three segments: *memorisation* occurs between epochs 0 and 1,500. The shift known as *grokking* unfolds from epoch 1,500 onward, lasting until around epoch 4,000. Beyond that point performance settles, which marks the *post-grokking* period. Large shifts in validation accuracy, along with changes in how representations are structured, define what we call a phase transition; we use “phase transition” descriptively, as an analogy of a thermodynamic phase transition.

We created the following event study for the original three seeds to show that the same geometric pattern appears at the transition. To compare the runs in the same chart, we indexed each run by the first checkpoint at which it reached a validation accuracy of 0.5. The result in Figure 1 shows that the dimensional collapse, validation accuracy transition, BW distances and coexact-led Hodge decomposition are all present at the generalisation transition across the three seeds.

3.2 Spatial Side: Velocity Hodge Decomposition

Figure 2 shows the DG smooth Hodge decomposition of the probe displacement field across all 50 consecutive checkpoint pairs. The character of the velocity changes at the grokking transition:

1. Memorisation phase (epochs 0–1,500): the velocity is coexact-dominated on average (mean $\sim 49\%$ coexact, $\sim 32\%$ exact, $\sim 19\%$ harmonic). Step 1,000 $\rightarrow$ 1,500 shows the strongest coexact fraction ($\sim 72\%$). This indicates curl-dominated rearrangement of the point cloud during memorisation. The initial step 0 $\rightarrow$ 500 still behaves differently, with elevated harmonic energy ($\sim 36\%$); this likely reflects large-scale compression from random initialisation and residual global structure in the harmonic component rather than direct topological evidence.

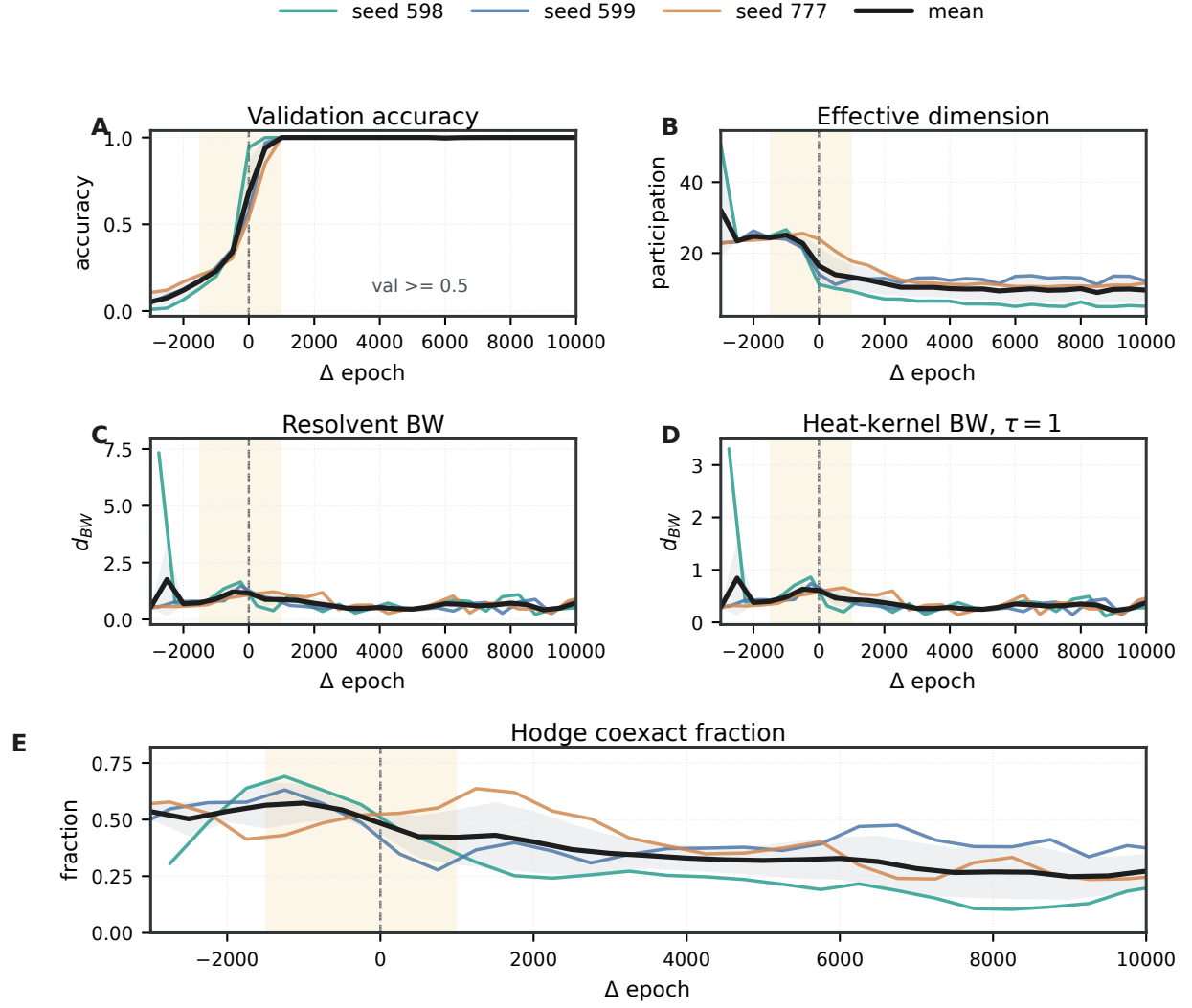


Figure 1: [Original three-seed event study](#) aligned by the first checkpoint where validation accuracy reaches 0.5. Panel A shows validation accuracy. Panel B shows effective dimension. Panel C shows consecutive resolvent BW distance. Panel D shows consecutive heat-kernel BW distance at $\tau = 1$. Panel E shows the Hodge coexact fraction.

2. Grokking transition (epochs 1,500–4,000): the exact component rises within the window as the Fourier structure emerges, but the phase remains coexact-leading on average ($\sim 33\%$ exact, $\sim 54\%$ coexact, $\sim 13\%$ harmonic). The transition is therefore a gradual rise of exact motion within a still-curl-dominated regime, with the actual flip happening only post-grokking in the baseline seed. Across eight seeds at the baseline setting, the transition means are 0.325 exact, 0.450 coexact, and 0.225 harmonic. Coexact exceeds exact on average, but transition coexact dominance is not invariant to seed or parameter choice.
3. Post-grokking (epochs 4,000+): the exact component leads on average (mean $\sim 69\%$ exact, $\sim 21\%$ coexact, $\sim 10\%$ harmonic). The representation continues to undergo directed refinement, while the harmonic term remains a residual global component of the finite DG decomposition rather than direct evidence for a topological cycle. Across eight seeds at the baseline setting, the post-grokking means are 0.473 exact, 0.331 coexact, and 0.196 harmonic. Exact exceeds coexact in six of eight baseline runs and in 44 of 56 seed-by-setting comparisons; the invariant result is the shift toward exact, not absolute exact dominance.

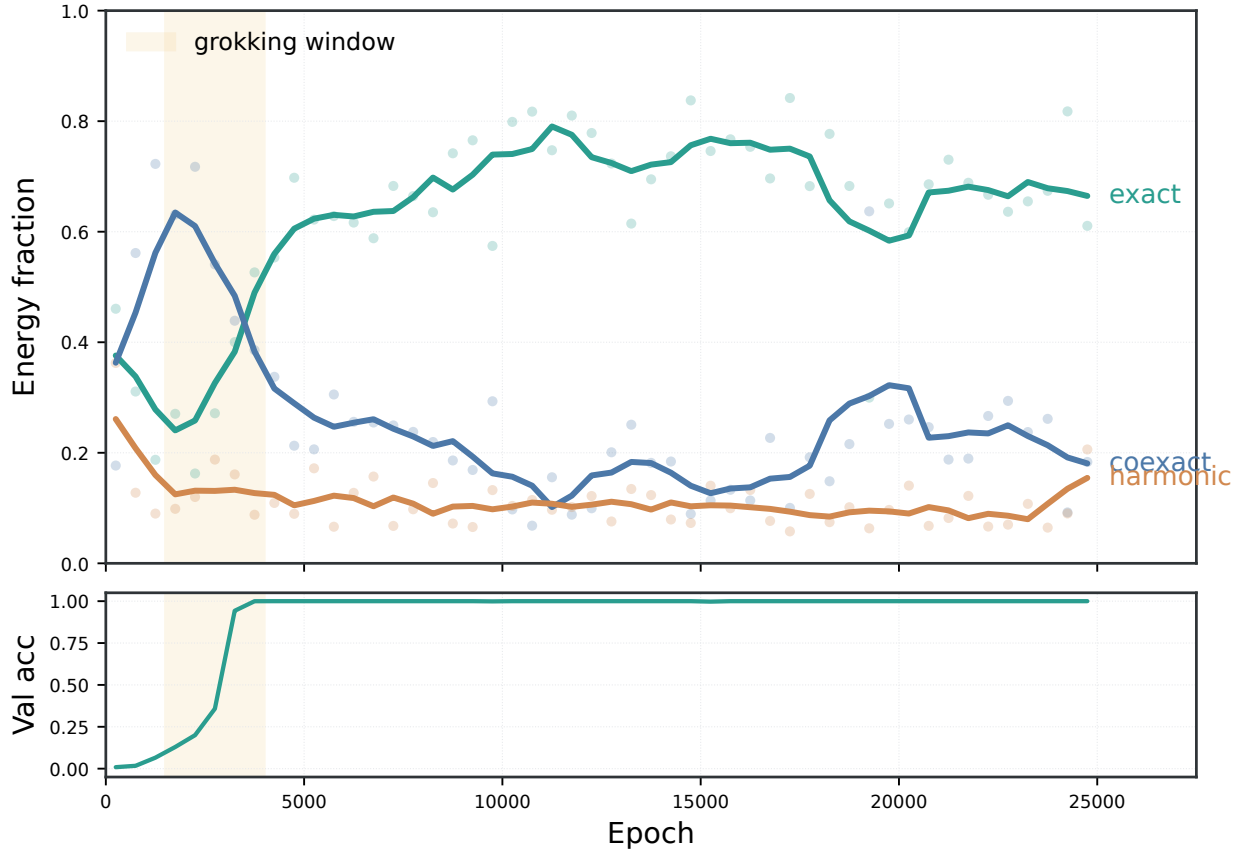


Figure 2: The displacement field changes character across grokking: early motion is curl-like, the transition is mixed, and post-grokking refinement is more gradient-like. The top panel shows exact, coexact, and harmonic energy fractions; the bottom panel shows validation accuracy.

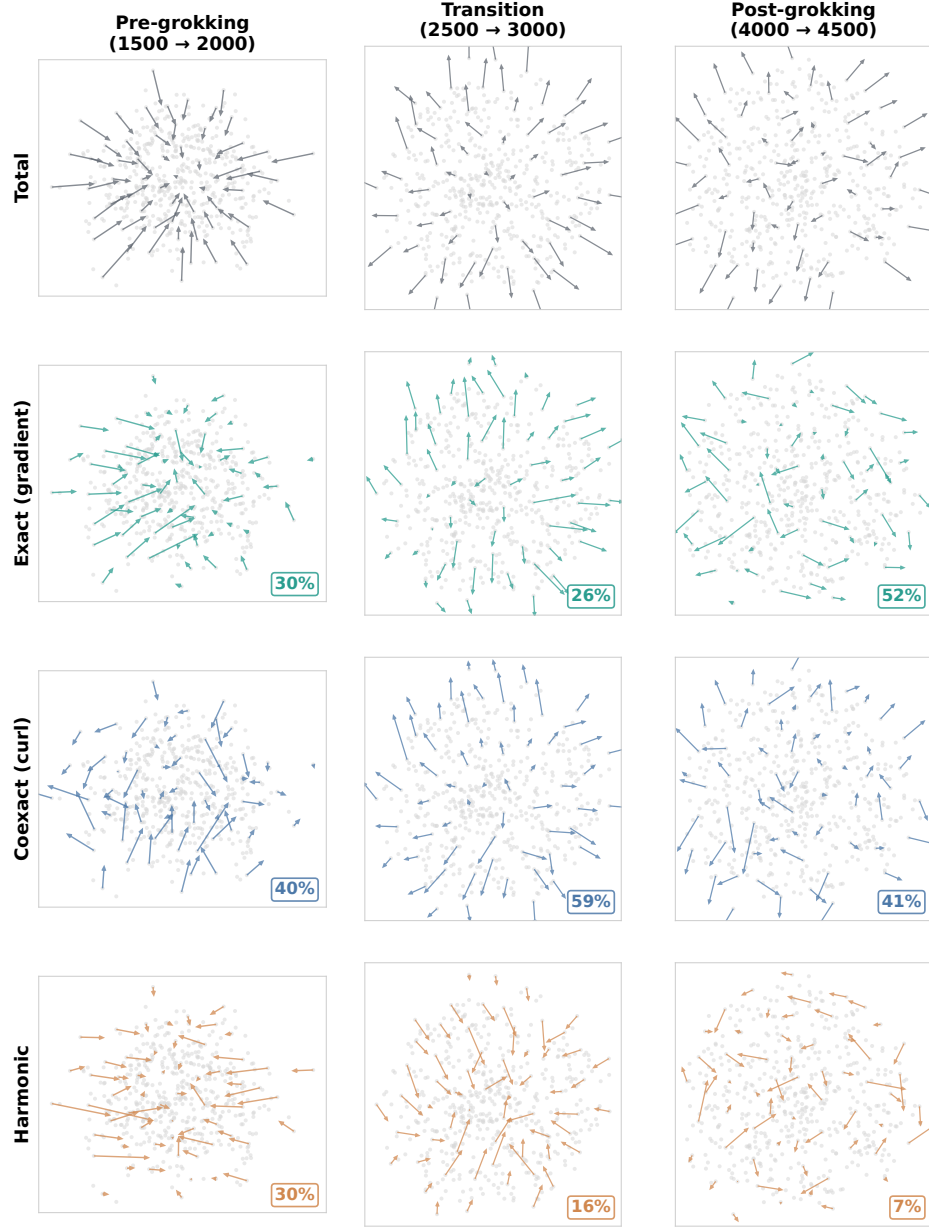


Figure 3: Decomposed velocity fields during grokking. Columns: pre-grokking (1,500–2,000), transition (2,500–3,000), and post-grokking (4,000–4,500). Rows: total velocity, exact (gradient), coexact (curl), and harmonic. Energy fractions are annotated.

3.2.1 Eight-Seed Robustness and Correspondence Null

Table 1 summarises the baseline Hodge composition using training seeds as the independent units.

Table 1: Baseline Hodge energy fractions across eight training seeds. Values are means of the seed-level phase means.

Phase	Exact	Coexact	Harmonic
Pre-grokking	0.342	0.488	0.170
Transition	0.325	0.450	0.225
Post-grokking	0.473	0.331	0.196

The reported -0.267 estimate is the baseline-setting contrast, not a pool over parameter settings. Within each seed we first average over its four probe subsets and selected checkpoint pairs, calculate $(\text{coexact} - \text{exact})_{\text{post}} - (\text{coexact} - \text{exact})_{\text{transition}}$, and then average the resulting eight seed contrasts. Its two-sided seed-level t interval is $[-0.400, -0.135]$. All 56/56 correlated fixed-window seed-by-setting changes are negative. After seed-relative phase alignment the baseline estimate is -0.360 ($[-0.487, -0.234]$), and all 56/56 sensitivity changes remain negative. Absolute post-grokking exact dominance occurs in 44/56 fixed-window comparisons and 50/56 seed-relative comparisons.

Table 2: Phase-definition sensitivity of the baseline contrast. The estimate and 95% interval use eight seed-level values; the two rightmost columns are descriptive counts of correlated sensitivity comparisons.

Phase definition	Mean change	95% seed-level t interval	Negative	Post exact-leading
Fixed epochs	-0.267	$[-0.400, -0.135]$	56/56	44/56
Seed-relative	-0.360	$[-0.487, -0.234]$	56/56	50/56

The correspondence null is based on 288 separately drawn permutations, one for each training-seed-probe-subset-checkpoint-pair baseline record. Each actual field is paired with its null at the same seed, subset, and checkpoint pair before within-seed averaging. Post-grokking, the actual-minus-null exact fraction is 0.249 (95% seed-level t interval $[0.155, 0.344]$), while the coexact difference is -0.208 ($[-0.311, -0.104]$).

The synthetic calibration assigns 93.2% of a gradient field to the exact component, 98.8% of a rotational field to the coexact component, and 94.2% of a discrete harmonic field to the harmonic component. Across the empirical sweep, the median between-subset standard deviations are 0.015 exact, 0.020 coexact, and 0.015 harmonic.

3.3 Global Corroborating Diagnostics

3.3.1 Effective Dimensionality

From the 500×512 activation matrix, prior to any PCA compression, we form the centred activation covariance C_t (we do this for the same reason centering is done in PCA) and derive the effective dimension as the participation ratio

$$d_{\text{eff}}(t) = \frac{(\text{tr } C_t)^2}{\text{tr}(C_t^2)}.$$

In other words, the ratio is the participation ratio of the covariance eigenvalues. After starting at 50.7, d_{eff} quickly decreases and is already down to ~ 24 by epoch 500. It keeps dropping until

reaching about 10 when grokking begins. Once past the transition, values flatten between 5 and 7, settling at ~ 6.4 by the final checkpoint (Figure 4). What remains is supported by just a few Fourier subspaces instead of spreading across many dimensions.

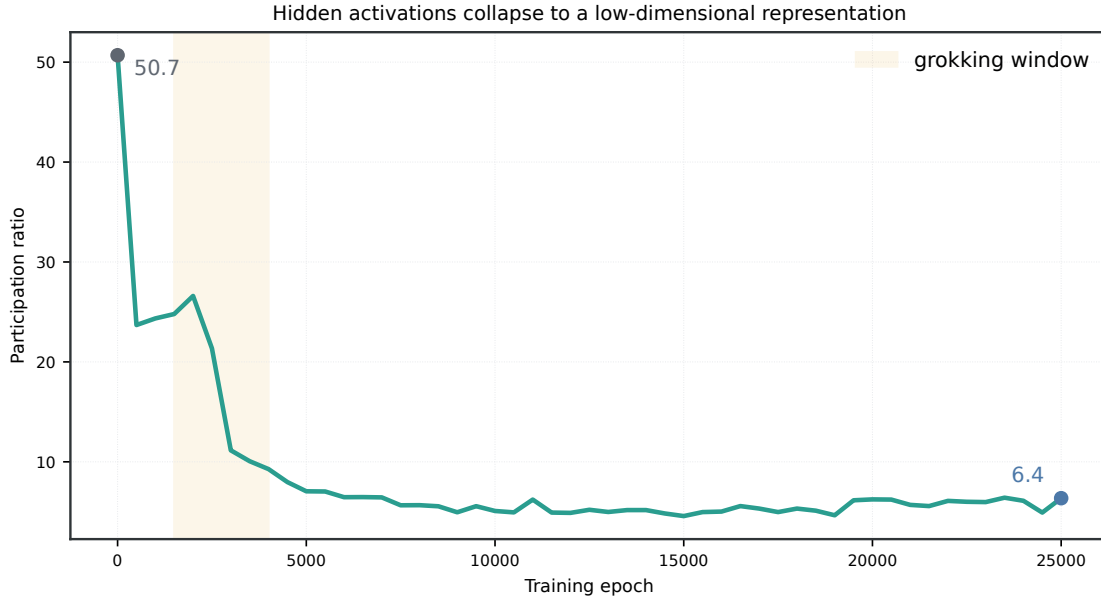


Figure 4: The hidden representation collapses from a diffuse high-dimensional cloud to a low-dimensional solution. The shaded window marks grokking; the curve shows the participation ratio falling from 50.7 at initialisation to ~ 6.4 by the final checkpoint. The earliest compression is largest, so this figure only shows dimensional collapse in general and is not yet a complete transition marker.

3.3.2 BW Distance Between Geometric States

The distance time series has three distinct features. The largest peak sits right at the start ($d_{\text{BW}} = 7.34$ at epochs 0–500), as the diffuse initial geometry gives way to learned structure. A second, smaller spike appears at the grokking transition ($d_{\text{BW}} = 1.36$ at 2,000–2,500, then 1.65 at 2,500–3,000). After that the consecutive distance settles into a much lower band, with mean $\bar{d}_{\text{BW}} \approx 0.56$ and occasional fluctuations up to 1.23.

4 Discussion

[This paper](#) reveals two distinct perspectives on how geometric structure develops. First, the operator-level lens applied to $L^\dagger$. Second, the Hodge approach on the velocity field $v_i(t) = x_i(t + \Delta t) - x_i(t)$. Both sides are available because vertex identities are stable across activation clouds. The operator-side reading reflects changes in the spectrum as geometry evolves, while the movement of individual points across the manifold becomes visible through spatial analysis. As the grokking transition happens, each of these two parts shows large changes at the same time. Across successive markers, the operator distance increases steeply while the velocity field shows that the nature of the velocity Hodge decomposition changes.

4.1 Summary of Grokking Findings

The spatial analysis showed that, in the baseline seed, the probe velocity evolves from a coexact-dominated (curl-like, $\sim 49\%$) regime during memorisation to an exact-dominated (directed, $\sim 69\%$) post-grokking regime, with the exact component rising during the transition window (epochs 1,500 to 4,000). On the operator side, the BW distance between consecutive resolvent covariances has a large initial spike, a secondary increase during the grokking transition, and then a much smaller range of values post-grokking. The effective dimensionality drops from 50.7 to 6.4. After generalisation, the consecutive resolvent BW distances, velocity Hodge, and effective dimensionality converge to stable values. We interpret this to mean that the representation has reached a geometrically stable state.

The eight-seed analysis narrows the Hodge interpretation. The transition-to-post change moves toward exact velocity in every seed and every tested parameter setting under both fixed and event-aligned phase definitions, while the absolute ordering of exact and coexact fractions varies. Statistical inference treats the eight training seeds as independent; the 56 within-seed setting comparisons are descriptive sensitivity checks. The correspondence null shows that the post-grokking exact enrichment depends on preserving point identity across checkpoints.

4.2 Grokking as Geometric Memory

The grokking experiment is a concrete case study for geometric memory [6]. The three phases of grokking align with three modes of geometric storage.

1. The first phase is pre-algorithmic memorisation (epochs 0–1,500). The representation forms a high-dimensional unordered arrangement that fits the training set but does not support generalisation. The BW analysis within this paper registers this as the largest geometric restructuring in the run. The effective dimensionality collapses, and the velocity Hodge decomposition shows coexact-led (curl-like) motion that increases until generalisation begins.
2. The second phase is geometric algorithmisation (epochs 1,500–4,000). The model shifts from storing individual training examples to storing the actual *algorithm* as a geometric configuration. This is the multi-frequency Fourier arrangement studied in this paper and in Nanda et al. [5]. It is supported by the effective dimensionality collapse and the velocity-field Hodge crossover (in the baseline seed).
3. The third and final phase is geometric refinement (epochs 4,000+). The algorithmic geometry has settled, the effective dimensionality plateaus at a low value, the BW distances drop into a low band, and the velocity is exact-dominated in the baseline seed as the network fine-tunes its Fourier embedding rather than reorganising its internal representations through rotation.

The key insight here is that memorisation and generalisation have distinct geometric signatures. The difference, therefore, is not between geometric and non-geometric storage. It is between what is stored geometrically: individual data points (memorisation) versus an algorithmic structure (generalisation). Grokking is the phase transition between these two modes of memory. The BW framework and Hodge decomposition employed in this paper delineate different aspects of the transition period, and together they illustrate algorithmic reorganisation in a neural network.

5 Conclusion

The primary contribution is a pointwise velocity-field Hodge decomposition across eight grokking runs. The effective dimension of the centred activation covariance collapses from 50.7 to 6.4 in the

baseline run, while the BW distance between successive resolvent operators shows a large initial spike, a smaller transition-time increase, and then remains in a smaller range once grokking is over. The completed Hodge sweep shows a transition-to-post shift toward exact velocity in all eight seeds and all 56 correlated seed-by-setting sensitivity comparisons. At the baseline setting, the seed-level mean change in coexact-minus-exact is -0.267 , and the correspondence null removes the post-grokking exact enrichment. Since absolute exact dominance occurs in 44/56 fixed-window comparisons, the supported conclusion is directional: generalisation is followed by movement toward a more exact velocity field, not by a universal component ordering in every run.

A Supplementary Geometric Diagnostics

A.1 Fourier-Supervised Projection

The ridge R^2 values reported below serve as in-sample diagnostics on a consistent set of probes. With $n < d$, the initialisation score rises artificially due to overfitting and should therefore not be read as pointing to real Fourier structure. Since the DG circular value tracks the best pair among numerous pairs of eigenfunctions, it serves better as a tool to spot top-performing pairs rather than persistent canonical coordinates.

Remark 1 (Fourier-supervised vs unsupervised projection). We found that vanilla PCA does not reliably recover the Fourier circle. Different weight initialisations orient the Fourier subspace differently inside $\mathbb{R}^{512}$ so the circle does not need to align with the leading PCs. We hence use a supervised projection via ridge regression (Appendix A.1). The ridge picks out the directions in activation space that encode a chosen Fourier frequency, regardless of how that subspace happens to sit in activation space. In the runs we checked, this projection recovers clean circles across the TransformerLens versions tested. This confirms that the Fourier structure is in the activations even when its PCA visibility depends on the seed we use.

We project a 500×512 activation block into 2D by fitting ridge regressions $w_{\cos}$ and $w_{\sin}$ that predict $\cos(2\pi k \cdot \ell/P)$ and $\sin(2\pi k \cdot \ell/P)$ from activations, then scatter plot $(w_{\cos}^\top x, w_{\sin}^\top x)$. The plotted axes are therefore Fourier-supervised axes and not PC1, PC2. The ridge picks out the 2D affine subspace of activation space most aligned with frequency k . Vanilla PCA picks the top variance subspace which, for the activations we use, does not cleanly isolate a single Fourier frequency (Remark 1). Finally, the ridge penalty $\alpha = 10$ matters here because the regression is in the $n < d$ regime (500 points in 512 dimensions), and the label $\ell_i = (a_i + b_i) \bmod P$ is the ground truth label.

We use the joint cos/sin R^2 as a diagnostic score, as it tells us how strongly the activations encode frequency. Figure 5 displays one high-scoring Fourier mode, $k = 38$ ($R^2 = 0.994$ at the final checkpoint); other criteria, as we discuss below, pick $k = 36$ or $k = 49$ instead.

The raw PCA diagnostic and the unsupervised DG scan below identify $k_{\text{dom}} = 36$ as the most stable frequency at epochs 2,500 and 3,000, with $k_{\text{dom}} = 38$ at the final checkpoint. This discrepancy reflects the multi-frequency nature of the solution (a torus, not a circle). Several nearby frequencies, including $k = 36, 38$ and 49 , all reach high post-grokking ridge scores. At initialisation the projection shows a diffuse cloud ($R^2 = 0.70$, largely reflecting in-sample overfitting in the $n < d$ regime), and by epoch 3,000 a clear circle has formed ($R^2 = 0.98$). Post-grokking the circle is well-defined with ordered colour strips ($R^2 \approx 0.99$).

A.2 Heat Kernel Multiscale Analysis

The complementary operator-level view replaces the resolvent covariance with the heat kernel

$$H_t(\tau) = e^{-\tau L_t},$$

Fourier circle emergence during grokking
(ridge projection onto cos/sin of frequency $k = 38$, coloured by Fourier phase)

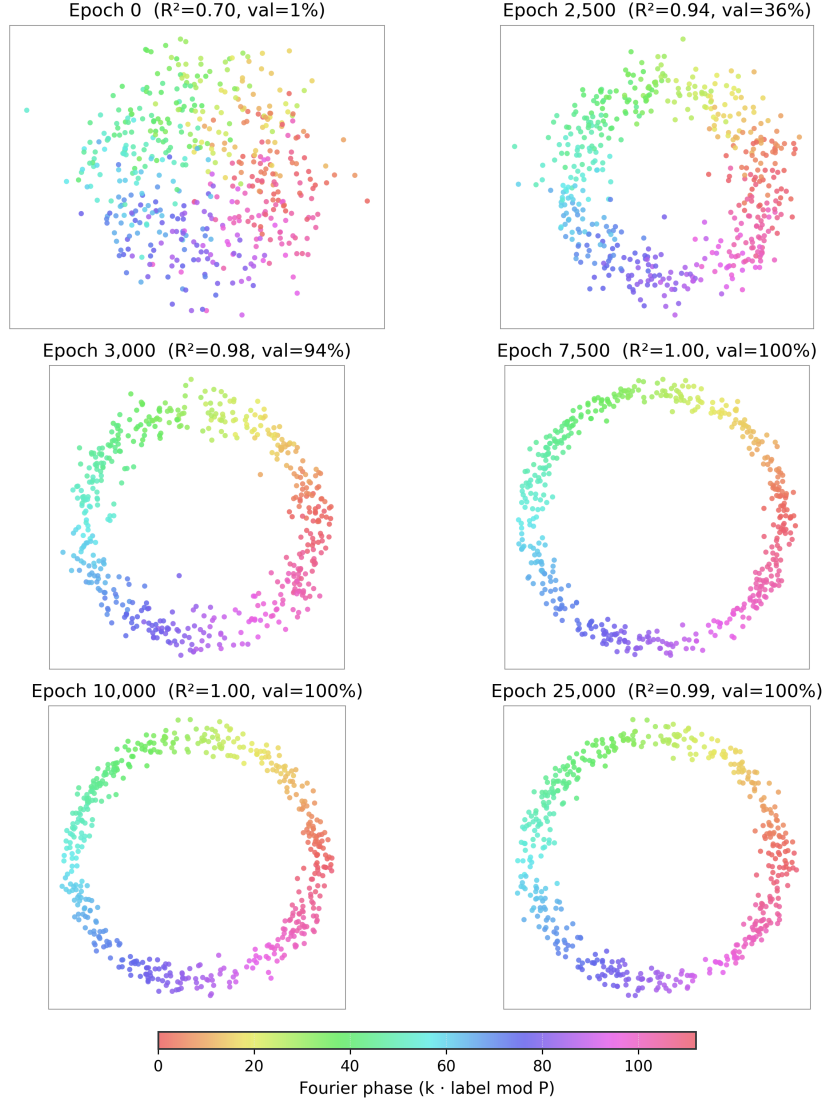


Figure 5: Fourier-supervised ridge projection of MLP hidden activations onto the strongest final-checkpoint Fourier mode, $k = 38$. Note that the plotted axes are ridge directions ($w_{\cos}^\top x$, $w_{\sin}^\top x$) rather than principal components (PC1, PC2). Points are coloured by Fourier phase $(k \cdot \ell) \bmod P$, which aligns the colour cycle with the displayed frequency. At initialisation the projection is noisy ($R^2 = 0.70$), which reflects ridge overfitting on the probes rather than Fourier structure. By epoch 3,000 a clean circle emerges ($R^2 = 0.98$) with smoothly ordered colour strips, confirming that a Fourier clock solution is present in the activations.

evaluated at three scales $\tau \in \{0.1, 1.0, 10.0\}$ (local, mesoscopic, global). For consistency with the resolvent analysis, the heat-kernel BW distances are computed on the same retained non-trivial spectral subspace unless otherwise stated. The scale parameter sets the diffusion time at which the kernel is read out, so each scale exposes a different rate of geometric reorganisation.

At different scales (τ set to 0.1, 1.0 and 10.0) the heat kernel delivers a clearer picture than the resolvent. While the resolvent captures broad trends, the heat kernel separates changes at distinct diffusion times. Right when training begins, the BW distance spikes at all scales and then quickly falls once that phase ends. The peak distances are 3.79 at the local scale, 3.31 at the mesoscopic scale, and 1.13 at the global scale.

Indeed the global peak is roughly one-third the local peak. Another small peak shows up across each scale near epochs 2,000 to 3,000, matching the shift into grokking. Post-grokking, the three curves settle to low baseline fluctuations.

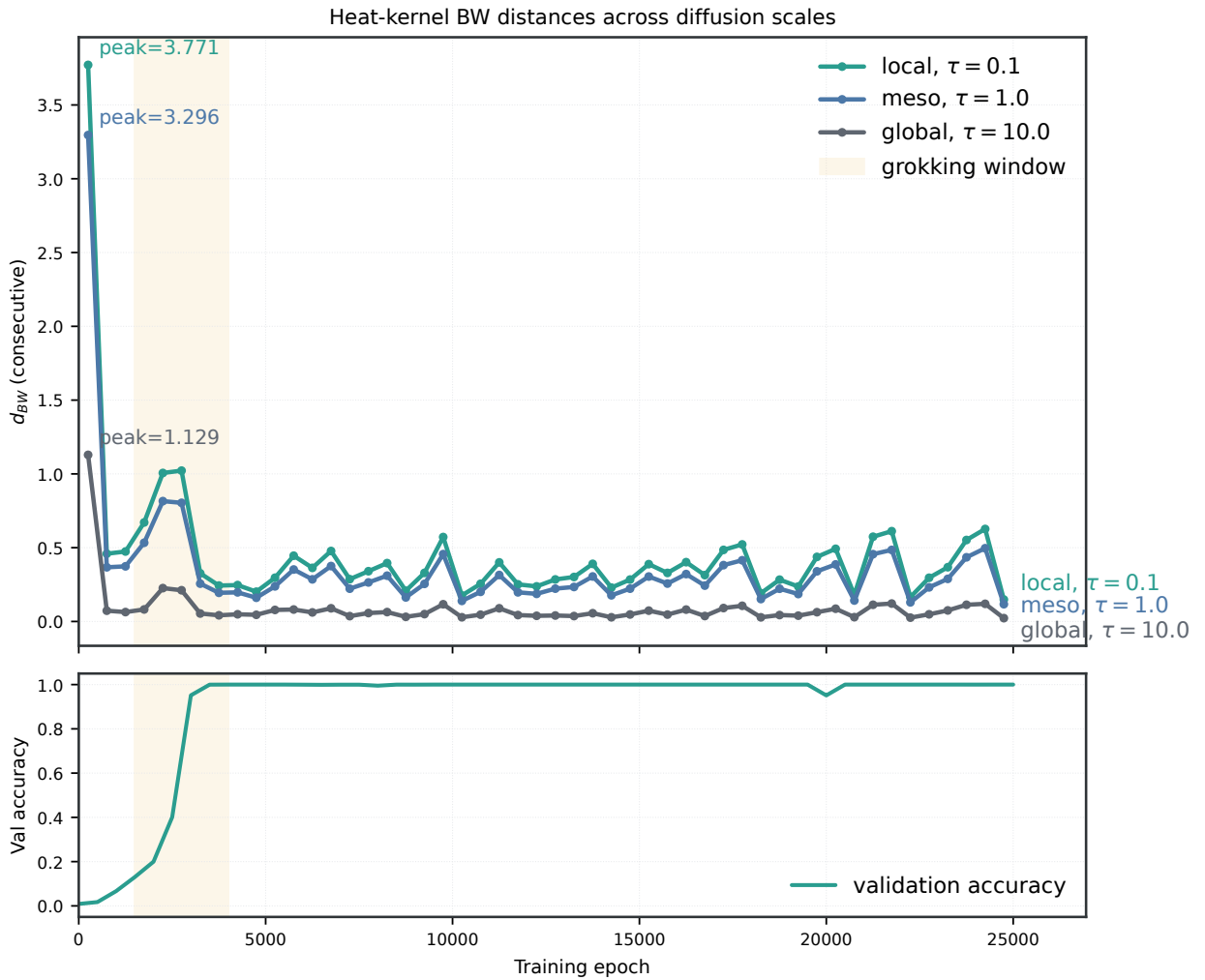


Figure 6: Heat-kernel BW shows the same operator change at the grokking transition across diffusion scales with the strongest locally and the weakest globally. The curves are consecutive BW distances between heat kernels at $\tau = 0.1, 1.0, 10.0$, with validation accuracy below. The shared shaded window makes the scale-specific increase comparable to the much larger initial restructuring.

B Claims, Evidence, and Limitations

Table 3: Claims, evidence, robustness checks, and limitations for the grokking case study.

Claim	Primary evidence	Robustness or diagnostic check	Remaining limitation
Grokking has a sharp training transition	Validation accuracy remains near chance until $\sim 1,500$ epochs, reaches 94.3% by 3,000, and reaches 100% by 4,000.	All geometric summaries are anchored to the same pre/transition/post timeline.	Eight trained seeds; 99% validation accuracy occurs between epochs 3,500 and 5,500.
Point motion shifts toward exact velocity after grokking	At the baseline setting, mean coexact-minus-exact changes by -0.267 from transition to post-grokking.	Eight independent seeds; four probe subsets; seven settings; 56/56 correlated directional sensitivity comparisons; seed-relative alignment; synthetic calibration; and a 288-permutation paired null.	Absolute post-grokking exact dominance occurs in 44/56 fixed-window comparisons; finite phase definitions and parameter grid.
Representation dimension collapses	Participation-ratio effective dimension falls from 50.7 at initialisation to 6.4 post-grokking.	Four 400-example probe subsets give $49.63 \pm 0.50 \rightarrow 6.40 \pm 0.03$, ratio 0.129 ± 0.002 . Three seeds give final/initial 0.126, 0.194, 0.250.	Probe-subset and three-seed checks.
Operator geometry restructures	Resolvent BW distance has a large initial spike ($d_{\text{BW}} = 7.34$), a transition-time burst (1.36, 1.65), and a lower post-grokking band.	Probe subsets preserve the pattern: initial 6.258 ± 0.050 , transition peak 1.662 ± 0.057 , post mean 0.531 ± 0.016 , transition/post 3.13 ± 0.09 . Three seeds give transition/post 2.92, 2.64, 1.48 (2.29, 1.84, 1.44 after accuracy alignment).	Graph, PCA, and reference-basis choices.
Local diffusion geometry changes at the transition	Heat-kernel BW curves show a secondary increase across local, meso, and global diffusion scales.	For probe subsets, transition/post ratios are 3.12 ± 0.11 ($\tau = 0.1$), 3.21 ± 0.11 ($\tau = 1$), and 3.36 ± 0.05 ($\tau = 10$); all four subsets have transition peak above post mean at every scale. Three seeds give $\tau = 1$ peak/post 2.99, 2.47, 1.52.	Only three diffusion scales are tested.
Algorithmic Fourier structure emerges	Fourier ridge projections and circular-coordinate scans recover post-grokking circular or multi-frequency structure.	The result is treated as diagnostic rather than a pre-registered coordinate claim.	In-sample ridge; best-pair circular scan; correlation varies in seeds 599 and 777.

Table 3 also identifies what would weaken our interpretation. The analysis on the operator side held under three independent training seeds for the dimensionality collapse and the BW spikes during the grokking transition. However, it could still lose its persuasiveness if those effects disappeared under held-out probe resampling. The Hodge decomposition now survives the completed eight-seed, probe, and one-at-a-time parameter sweeps in its directional claim. The circular-coordinate diagnostics remain supplementary and do not receive the same robustness claim.

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